

# Metricas implementadas STLC/ISTQB

Este documento muestra solo las formulas que se ejecutan en el codigo de la plataforma.

Regla comun usada en reportes:

| Funcion                    | Archivo                     | Formula                                               |
|----------------------------|-----------------------------|-------------------------------------------------------|
| `_percentage(part, total)` | `src/apps/reports/views.py` | `si total > 0: round((part / total) * 100); si no: 0` |
| `_ratio(part, total)`      | `src/apps/reports/views.py` | `si total > 0: round(part / total, 2); si no: 0`      |

## Tabla de formulas

| #  | Metrica                            | Formula ejecutada                                                                                                 | Archivo / funcion                                                                                                                                                                     |
|----|------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Cobertura de requisitos            | `coverage = round((covered_requirements / total_requirements) * 100)`                                             | `src/apps/reports/views.py` -> `_build_coverage_content`; `src/apps/traceability/views.py` -> `traceability_matrix_view`; `src/apps/requirements/views.py` -> `requirement_list_view` |
| 2  | Requisito cubierto                 | `coverage = 100 si linked_cases > 0; si no coverage = 0`                                                          | `src/apps/requirements/views.py` -> `requirement_list_view`; `src/apps/traceability/views.py` -> `traceability_matrix_view`                                                           |
| 3  | Casos vinculados a un requisito    | `linked_cases = max(direct_cases, traced_cases)`                                                                  | `src/apps/requirements/views.py` -> `requirement_list_view`                                                                                                                           |
| 4  | Requisitos sin cobertura           | `uncovered_requirements = total_requirements - covered_requirements`                                              | `src/apps/reports/views.py` -> `_build_coverage_content`                                                                                                                              |
| 5  | Casos de prueba vinculados         | `linked_test_cases = count(TC donde TC.requirement != null OR TC.traceability_links != null)`                     | `src/apps/reports/views.py` -> `_build_coverage_content`                                                                                                                              |
| 6  | Requisitos automatizados           | `automated_requirements = count(requisitos con casos que tienen automated_rules.is_active = True)`                | `src/apps/reports/views.py` -> `_build_coverage_content`                                                                                                                              |
| 7  | Requisitos solo manuales           | `manual_only_requirements = max(covered_requirements - automated_requirements, 0)`                                | `src/apps/reports/views.py` -> `_build_coverage_content`                                                                                                                              |
| 8  | Avance de ejecucion                | `execution_progress = round((executed_cases / total_test_cases) * 100)`                                           | `src/apps/reports/views.py` -> `_build_execution_content`; `src/apps/reports/views.py` -> `_build_summary_content`                                                                    |
| 9  | Casos ejecutados                   | `executed_cases = count(distinct test_case donde execution.result != NOT_RUN)`                                    | `src/apps/reports/views.py` -> `_build_execution_content`, `_build_summary_content`, `_build_defects_content`, `_build_final_content`                                                 |
| 10 | Progreso del proyecto              | `progress = round((passed_cases / total_cases) * 100)`                                                            | `src/apps/dashboard/views/dashboard_views.py` -> `get_project_progress`; `src/apps/projects/views.py` -> `build_project_card`, `project_detail_view`                                  |
| 11 | Ejecuciones aprobadas              | `passed = count(executions donde result = PASSED)`                                                                | `src/apps/reports/views.py` -> `_build_execution_content`                                                                                                                             |
| 12 | Ejecuciones fallidas               | `failed = count(executions donde result = FAILED)`                                                                | `src/apps/reports/views.py` -> `_build_execution_content`                                                                                                                             |
| 13 | Ejecuciones bloqueadas             | `blocked = count(executions donde result = BLOCKED)`                                                              | `src/apps/reports/views.py` -> `_build_execution_content`                                                                                                                             |
| 14 | Ejecuciones con error              | `errors = count(executions donde result = ERROR)`                                                                 | `src/apps/reports/views.py` -> `_build_execution_content`                                                                                                                             |
| 15 | Ejecuciones no realizadas          | `not_run = count(executions donde result = NOT_RUN)`                                                              | `src/apps/reports/views.py` -> `_build_execution_content`                                                                                                                             |
| 16 | Porcentaje por estado de ejecucion | `percent_status = round((count_status / total_executions) * 100)`                                                 | `src/apps/reports/views.py` -> `_execution_status_distribution`                                                                                                                       |
| 17 | Tasa de aprobacion                 | `success_rate = round((passed / (total_executions - not_run)) * 100)`                                             | `src/apps/reports/views.py` -> `_build_execution_content`                                                                                                                             |
| 18 | Exito historico de un caso         | `success_percent = round((passed_count / execution_total) * 100)`                                                 | `src/apps/executions/views.py` -> `execution_workspace_view`, `execution_history_view`                                                                                                |
| 19 | Resultado manual por pasos         | `si algun paso = FAILED -> FAILED; si no, si algun paso = BLOCKED -> BLOCKED; si no -> PASSED`                    | `src/apps/executions/views.py` -> `aggregate_step_result`                                                                                                                             |
| 20 | Resultado semi-automatizado        | `FAILED > ERROR > BLOCKED > PASSED; si no hay reglas validas -> BLOCKED`                                          | `src/apps/executions/services/automated_runner.py` -> `aggregate_automated_status`                                                                                                    |
| 21 | Duracion semi-automatizada         | `duration_seconds = Decimal((finished_at - started_at) / 1 segundo).quantize(0.001)`                              | `src/apps/executions/services/automated_runner.py` -> `run_automated_execution`                                                                                                       |
| 22 | Ejecuciones con evidencia          | `executions_with_evidence = count(distinct execution donde evidence != "" OR automated_results.screenshot != "")` | `src/apps/reports/views.py` -> `_build_execution_content`                                                                                                                             |
| 23 | Tasa de evidencia                  | `evidence_rate = round((executions_with_evidence / total_executions) * 100)`                                      | `src/apps/reports/views.py` -> `_build_final_content`                                                                                                                                 |

| #  | Metrica                             | Formula ejecutada                                                                                                                  | Archivo / funcion                                                                                                 |
|----|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| 24 | Revision docente                    | `reviewed_executions = validated + rejected + needs_fix`                                                                           | `src/apps/reports/views.py` -> `_build_final_content`                                                             |
| 25 | Tasa de revision docente            | `review_rate = round((reviewed_executions / total_executions) * 100)`                                                              | `src/apps/reports/views.py` -> `_build_final_content`                                                             |
| 26 | Requisitos ejecutados               | `executed_requirements = count(requisito donde existe latest_execution en sus casos)`                                              | `src/apps/reports/views.py` -> `_requirement_execution_metrics`                                                   |
| 27 | Requisito fallido                   | `failed_requirement = existe latest_result = FAILED`                                                                               | `src/apps/reports/views.py` -> `_requirement_execution_metrics`                                                   |
| 28 | Requisito bloqueado                 | `blocked_requirement = existe latest_result en {BLOCKED, ERROR}`                                                                   | `src/apps/reports/views.py` -> `_requirement_execution_metrics`                                                   |
| 29 | Requisito aprobado                  | `passed_requirement = todos los latest_result = PASSED`                                                                            | `src/apps/reports/views.py` -> `_requirement_execution_metrics`                                                   |
| 30 | Tasa de ejecucion de requisitos     | `requirement_execution_rate = round((executed_requirements / total_requirements) * 100)`                                           | `src/apps/reports/views.py` -> `_build_final_content`                                                             |
| 31 | Defectos totales                    | `defects = count(defects)`                                                                                                         | `src/apps/reports/views.py` -> `_build_defects_content`                                                           |
| 32 | Defectos abiertos                   | `open_defects = count(defects donde status = OPEN)`                                                                                | `src/apps/reports/views.py` -> `_build_defects_content`                                                           |
| 33 | Defectos en progreso                | `in_progress_defects = count(defects donde status = IN_PROGRESS)`                                                                  | `src/apps/reports/views.py` -> `_build_defects_content`                                                           |
| 34 | Defectos en analisis                | `analysis_defects = count(defects donde status = ANALYSIS)`                                                                        | `src/apps/reports/views.py` -> `_build_defects_content`                                                           |
| 35 | Defectos pendientes de confirmacion | `pending_confirmation_defects = count(defects donde status = PENDING_CONFIRMATION)`                                                | `src/apps/reports/views.py` -> `_build_defects_content`                                                           |
| 36 | Defectos cerrados                   | `closed_defects = count(defects donde status = CLOSED)`                                                                            | `src/apps/reports/views.py` -> `_build_defects_content`                                                           |
| 37 | Defectos por severidad              | `defects_severity = count(defects donde severity = CRITICAL/HIGH/MEDIUM/LOW)`                                                      | `src/apps/reports/views.py` -> `_build_defects_content`, `_defect_severity_rows`                                  |
| 38 | Defectos por estado                 | `defects_status = count(defects donde status = estado)`                                                                            | `src/apps/reports/views.py` -> `_defect_status_rows`                                                              |
| 39 | Densidad de defectos                | `defect_density = round(total_defects / executed_cases, 2)`                                                                        | `src/apps/reports/views.py` -> `_build_defects_content`, `_build_execution_content`, `_build_summary_content`     |
| 40 | Defectos trazados a ejecucion       | `defects_with_execution = count(defects donde execution != null)`                                                                  | `src/apps/reports/views.py` -> `_build_defects_content`, `_build_final_content`                                   |
| 41 | Tasa de trazabilidad de defectos    | `si total_defects > 0: round((defects_with_execution / total_defects) * 100); si no: 100`                                          | `src/apps/reports/views.py` -> `_build_final_content`                                                             |
| 42 | Tasa de deteccion de defectos       | `detection_rate = round((total_defects / total_executions) * 100)`                                                                 | `src/apps/reports/views.py` -> `_build_final_content`                                                             |
| 43 | Tasa de correccion de defectos      | `correction_rate = round((closed_defects / total_defects) * 100)`                                                                  | `src/apps/reports/views.py` -> `_build_final_content`                                                             |
| 44 | Riesgos altos                       | `high_risks = count(risk donde risk.risk_level = "Alto")`                                                                          | `src/apps/reports/views.py` -> `_high_risk_count`; `src/apps/traceability/views.py` -> `traceability_matrix_view` |
| 45 | Nivel de riesgo alto                | `Alto = (probability = HIGH AND impact in {MEDIUM, HIGH}) OR (impact = HIGH AND probability in {MEDIUM, HIGH})`                    | `src/apps/incidents/models.py` -> `risk_level`                                                                    |
| 46 | Nivel de riesgo bajo                | `Bajo = (probability = LOW AND impact in {LOW, MEDIUM}) OR (probability = MEDIUM AND impact = LOW)`                                | `src/apps/incidents/models.py` -> `risk_level`                                                                    |
| 47 | Nivel de riesgo medio               | `Medio = cualquier combinacion que no sea Alto ni Bajo`                                                                            | `src/apps/incidents/models.py` -> `risk_level`                                                                    |
| 48 | Matriz de riesgos                   | `risk_cell = count(risks donde probability = p AND impact = i)`                                                                    | `src/apps/reports/views.py` -> `_risk_matrix_rows`                                                                |
| 49 | Indice global de trazabilidad       | `traceability_index = round((coverage + requirement_execution_rate + evidence_rate + review_rate + defect_traceability_rate) / 5)` | `src/apps/reports/views.py` -> `_build_final_content`                                                             |
| 50 | Progreso de fase                    | `phase_progress = round((completed_checks / total_checks) * 100)`                                                                  | `src/apps/phases/views.py` -> `phase_criteria_status`                                                             |
| 51 | Tareas pendientes de fase           | `pending_tasks = total_checks - completed_checks`                                                                                  | `src/apps/phases/views.py` -> `phase_criteria_status`                                                             |
| 52 | Progreso general de fases           | `general_progress = round(sum(phase.progress) / phase_count)`                                                                      | `src/apps/phases/views.py` -> `phase_list_view`                                                                   |
| 53 | Fase completable                    | `can_complete = total_checks > 0 AND pending_tasks = 0`                                                                            | `src/apps/phases/views.py` -> `phase_criteria_status`                                                             |
| 54 | Criterio de cobertura minima        | `coverage >= plan.minimum_coverage_percentage`                                                                                     | `src/apps/reports/views.py` -> `_exit_criteria`                                                                   |
| 55 | Criterio de aprobacion minima       | `success_rate >= plan.minimum_pass_percentage`                                                                                     | `src/apps/reports/views.py` -> `_exit_criteria`                                                                   |
| 56 | Criterio de defectos criticos       | `critical_defects <= plan.maximum_critical_defects`                                                                                | `src/apps/reports/views.py` -> `_exit_criteria`                                                                   |

| #  | Metrica                       | Formula ejecutada                                                                                                           | Archivo / funcion                                                       |
|----|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| 57 | Criterios de salida aprobados | `exit_passed = criterio_coverage AND criterio_success AND criterio_critical_defects`                                        | `src/apps/reports/views.py` -> `_exit_criterio`                         |
| 58 | Veredicto aprobado            | `APROBADO = exit_passed AND critical_defects = 0 AND open_defects = 0 AND high_risks = 0 AND pending_review_executions = 0` | `src/apps/reports/views.py` -> `_exit_criterio`, `_build_final_content` |
| 59 | Veredicto con observaciones   | `APROBADO CON OBSERVACIONES = exit_passed OR coverage >= minimum_coverage_percentage`                                       | `src/apps/reports/views.py` -> `_exit_criterio`, `_build_final_content` |
| 60 | Veredicto no aprobado         | `NO APROBADO = no se cumplen las condiciones anteriores`                                                                    | `src/apps/reports/views.py` -> `_exit_criterio`                         |

Variables principales

| Variable                                           | Significado en el codigo                                     |
|----------------------------------------------------|--------------------------------------------------------------|
| `total_requirements`                               | Total de requisitos del proyecto                             |
| `covered_requirements`                             | Requisitos con al menos un caso de prueba directo o trazado  |
| `direct_cases`                                     | Casos asociados por `TestCase.requirement`                   |
| `traced_cases`                                     | Casos asociados por `TraceabilityLink`                       |
| `total_test_cases` / `test_cases.count()`          | Total de casos de prueba                                     |
| `executed_cases`                                   | Casos con al menos una ejecucion distinta de `NOT_RUN`       |
| `total_executions`                                 | Total de ejecuciones registradas                             |
| `passed`, `failed`, `blocked`, `errors`, `not_run` | Conteos por resultado de ejecucion                           |
| `executions_with_evidence`                         | Ejecuciones con evidencia o captura automatizada             |
| `reviewed_executions`                              | Ejecuciones validadas, rechazadas o marcadas para correccion |
| `total_defects` / `defects.count()`                | Total de defectos                                            |
| `closed_defects`                                   | Defectos cerrados                                            |
| `critical_defects`                                 | Defectos con severidad critica                               |
| `high_risks`                                       | Riesgos cuyo nivel calculado es Alto                         |
| `completed_checks`                                 | Criterios completados en una fase                            |
| `total_checks`                                     | Total de criterios evaluados en una fase                     |

Formulas compuestas clave

Cobertura de requisitos

covered\_requirements = requisitos con caso directo OR enlace de trazabilidad<br/>coverage = round((covered\_requirements / total\_requirements) \* 100)

Avance de ejecucion

executed\_cases = count(distinct test\_case donde result != NOT\_RUN)<br/>execution\_progress = round((executed\_cases / total\_test\_cases) \* 100)

Tasa de aprobacion

```
executed = total_executions - not_run<br/>success_rate = round((passed / executed) * 100)
```

**Densidad de defectos**

```
defect_density = round(total_defects / executed_cases, 2)
```

**Indice global de trazabilidad**

```
traceability_index =<br/>round((coverage + requirement_execution_rate + evidence_rate + review_rate + defect_traceability_rate) / 5)
```

**Criterios de salida**

```
exit_passed =<br/>(coverage >= minimum_coverage_percentage)<br/>AND (success_rate >= minimum_pass_percentage)<br/>AND (critical_defects <= maximum_critical_defects)
```